SJCET PALAI

AGRICONNECT

DBMS MINI PROJECT
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OCTOBER 2026

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- Testing & Results — 43 cases, lifecycle, Aadhaar, Playwright
- SDGs, POs/PSOs & Conclusion — impact, outcomes, references

INTRODUCTION

- Web marketplace connecting farmers directly with buyers
- PostgreSQL/Supabase database with 10 tables
- Trigger updates stock and marks listings sold out
- Buyers search, filter and place orders
- Rule-based Crop Advisor; Aadhaar stored as salted SHA-256
- 10 views and 9 stored functions power reports



PROBLEM STATEMENT

- Farmers depend on intermediaries and lose margin
- Prices and demand are scattered across informal channels
- Orders and payments are tracked manually
- Buyers cannot compare produce reliably
- Sales and inventory reporting is slow



OBJECTIVES

- Build a direct farmer-to-buyer marketplace
- Enable listing, discovery and ordering in one platform
- Track every order through a status timeline
- Recommend crops from soil, weather and season inputs
- Surface relevant government schemes
- Generate sales, buyer, inventory and low-stock reports

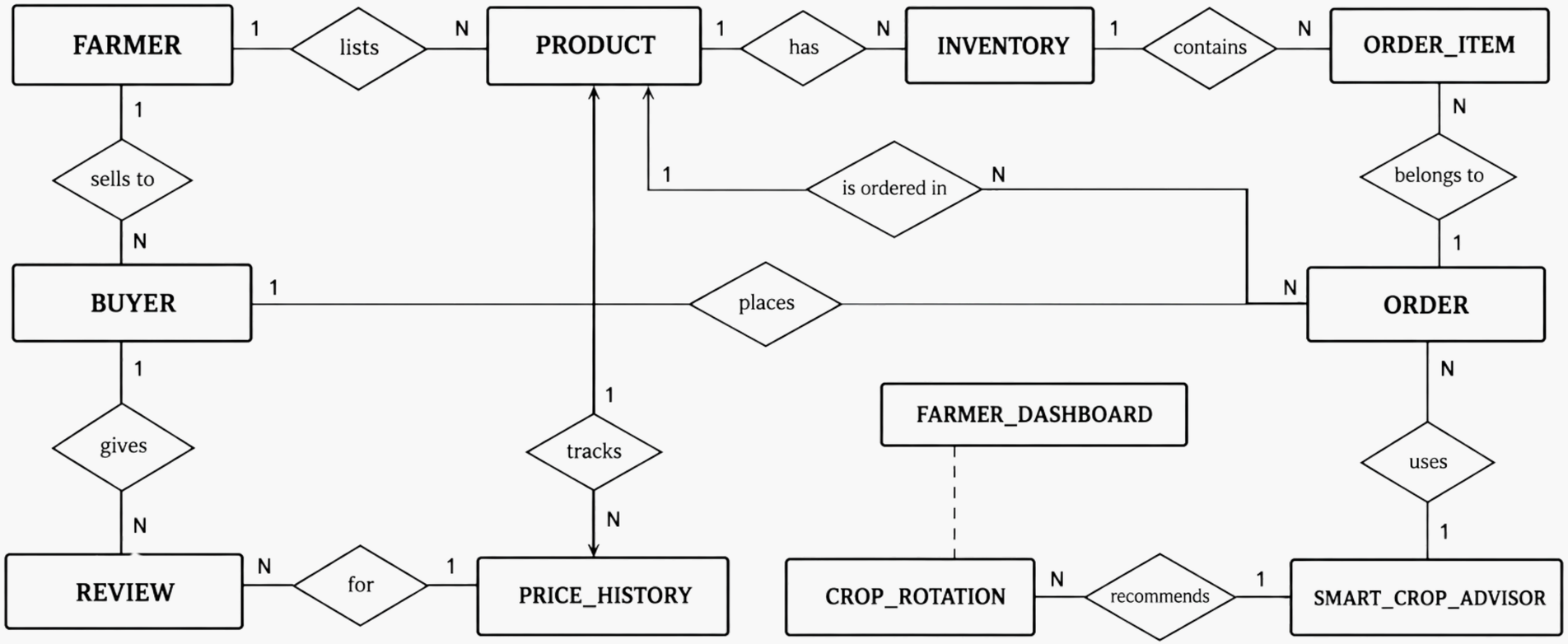


LITERATURE SURVEY

- eNAM (National Agriculture Market): Provides a unified agricultural market but can be complex for small-scale farmers.
- ONDC for Agriculture: Offers opportunities for digital commerce but lacks some integrated advisory and scheme-related features.
- Existing Marketplaces: Primarily focus on e-commerce rather than specialized agricultural workflows.
- Research Gap: A need exists for a simple, role-based platform that combines produce trading, crop advisory, government schemes, and order management.

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ER DIAGRAM



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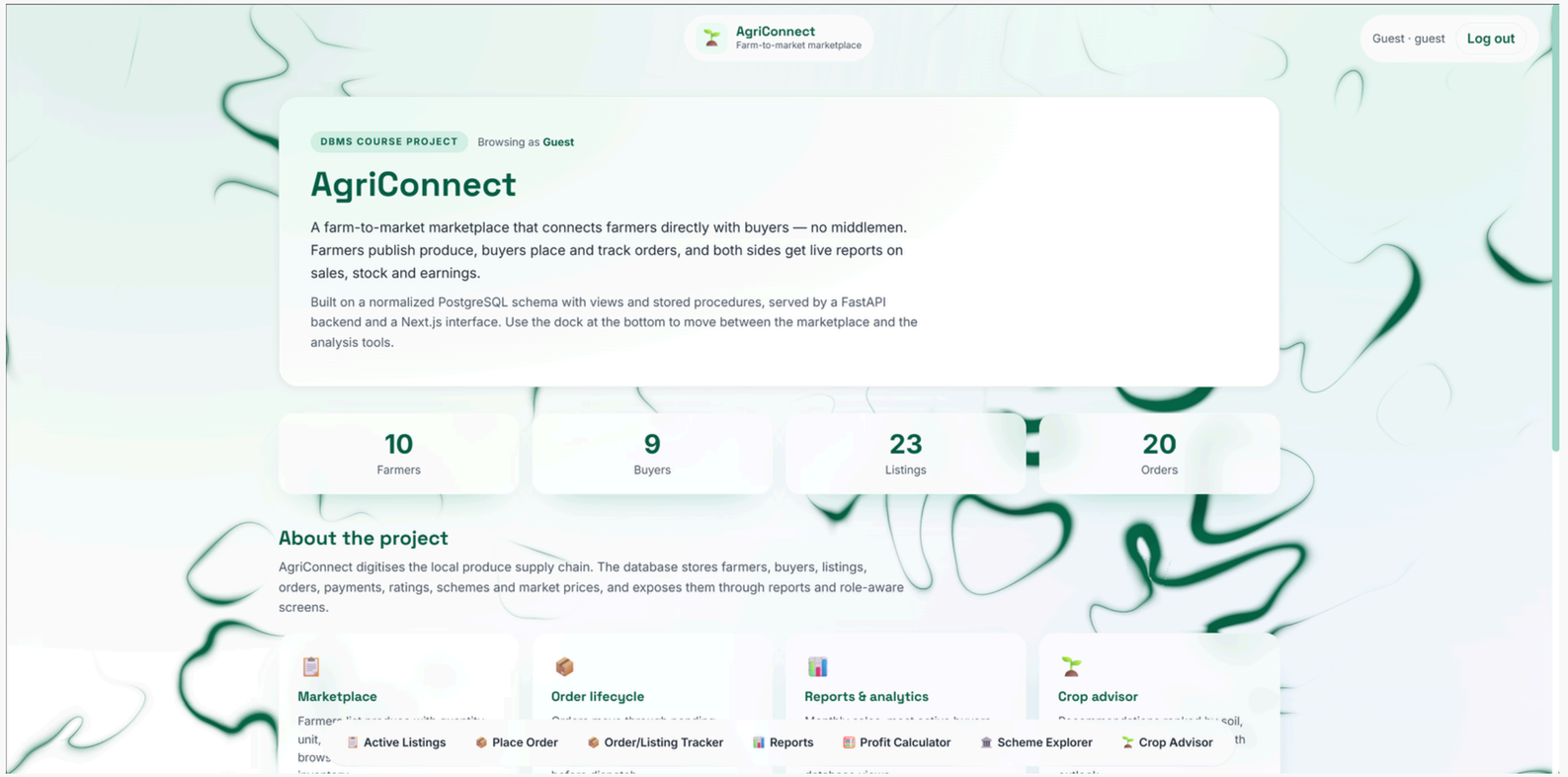
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TECHNOLOGY STACK + SYSTEM ARCHITECTURE



- Frontend: Next.js 16, React 19, TypeScript
- Backend: FastAPI, Uvicorn, Python 3.13
- Database: Supabase PostgreSQL
- Auth: Aadhaar login via RPC; farmer and buyer roles
- Hosting: NAS + secure tunnel

WEBSITE INTERFACE



- Use current dashboard screenshot
- Farmer dashboard with live counts
- Active listings with search and filters
- Order placement and status tracking
- Rule-based Crop Advisor and scheme explorer
- Reports and profit calculator

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MODULES AND FEATURES

- Marketplace: listings, search, filters and CSV export
- Orders: placement, tracking, status timeline and cancellation
- Crop Advisor: soil, weather and season inputs
- Farmer tools: profit calculator, schemes and reports
- Auth: Aadhaar demo login; farmer, buyer and guest roles

DATABASE DESIGN HIGHLIGHTS

Core Tables:

- farmers
- buyers
- listings
- orders
- payments
- ratings
- schemes
- price_history
- crops
- order_status_history

Important Database Views:

- active_listings_view
- order_history_view
- order_status_audit_view
- inventory_view
- low_stock_view
- monthly_sales_summary_view
- most_active_buyers_view
- farmer_avg_rating_view

Key RPC Functions:

- aadhaar_farmer_login
- aadhaar_buyer_login
- recommend_crops
- update_order_status
- get_farmer_revenue
- get_most_active_buyers
- get_farmer_ratings
- set_farmer_aadhaar
- set_buyer_aadhaar

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UI DESIGN AND IMPLEMENTATION

- Emerald glassmorphism; Space Grotesk and Inter
- Fixed top dock and floating bottom navigation
- Translucent cards, gradient accents and soft shadows
- Responsive at 320, 360 and 390 px
- Verified with Playwright for layout and console errors



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TESTING AND RESULTS

Testing Performed:

- 43 functional test cases; all passed
- Role access tested for farmer, buyer and guest
- Order lifecycle: pending → confirmed → delivered → paid
- Stock trigger and low-stock checks passed
- Aadhaar validation and negative cases passed
- Playwright checks at 320, 360 and 390 px

Results:

- All core features function as expected.
- Role-based access restrictions work as intended.
- Stock updates automatically when orders are placed.
- Crop recommendations respond to the provided soil inputs.
- UI testing completed with zero runtime exceptions.

CONCLUSION, FUTURE SCOPE

- Direct marketplace improves price transparency
- One platform for trade, advisory, schemes and reports
- Current scope: rule-based advice and tracked payments only
- Future: multilingual support and mobile app
- Future: UPI/Razorpay, notifications and logistics
- Future: ML-based price prediction

AGRICONNECT AND THE SDG'S

- SDG 2: direct market access and better farmer income
- SDG 8: digital entrepreneurship and revenue tracking
- SDG 9: low-cost digital marketplace infrastructure
- SDG 10: reduces information asymmetry
- **Supporting: SDG 1, 12 and 13**

PO/PSO MAPPING

- PO1, PO2, PO3: engineering knowledge, problem analysis, solution design
- PO5, PO6: modern tools and societal impact
- PO8: ethics, access control and Aadhaar privacy
- PSO1: analyse and design database-backed systems
- PSO2: use modern tools to design and communicate

REFERENCES

- Next.js, React and FastAPI documentation
- PostgreSQL, Supabase and Playwright documentation
- eNAM and ONDC official sources
- Department of Agriculture & Farmers Welfare
- Open-Meteo and Nominatim
- (plus eNAM/ONDC official sources from literature slide)

